 

# IBRI, Oman

WMO: **412520**

Lat: **23.1954N** Lon: **56.4295E** Elev: **327** StdP: **97.46** Time Zone: **4.00 (E04)** Period: **04-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **11.4** | **12.8** | **-7.8** | **2.0** | **28.4** | **-5.7** | **2.4** | **28.6** | **10.9** | **21.5** | **10.1** | **22.8** | **2.9** | **310** | **0.468** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **12.1** | **46.0** | **20.9** | **45.1** | **21.0** | **44.1** | **21.1** | **25.5** | **36.1** | **24.9** | **36.4** | **24.4** | **36.2** | **5.1** | **310** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **22.7** | **18.2** | **30.7** | **21.8** | **17.1** | **30.9** | **21.0** | **16.3** | **31.1** | **80.5** | **36.5** | **77.9** | **36.2** | **75.9** | **36.4** | **29.0** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **10.6** | **9.2** | **8.0** | DB | **8.6** | **47.9** | **1.4** | **0.6** | **7.6** | **48.3** | **6.7** | **48.7** | **5.9** | **49.0** | **4.8** | **49.4** | (4) |
| (5) | | | | WB | **5.4** | **26.8** | **1.4** | **1.1** | **4.4** | **27.6** | **3.5** | **28.3** | **2.7** | **28.9** | **1.7** | **29.7** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | **Temperatures, Degree-Days and Degree-Hours** | | DBAvg | **29.8** | **19.9** | **22.0** | **26.0** | **30.3** | **34.9** | **37.0** | **37.9** | **36.8** | **34.7** | **31.1** | **25.5** | **21.5** | (6) |
| (7) | | | DBStd | **6.75** | **2.24** | **3.36** | **3.01** | **3.11** | **2.22** | **2.07** | **1.90** | **2.22** | **1.98** | **1.93** | **2.22** | **2.17** | (7) |
| (8) | | | HDD10.0 | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | (8) |
| (9) | | | HDD18.3 | **18** | **10** | **6** | **1** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **0** | **2** | (9) |
| (10) | | | CDD10.0 | **7244** | **308** | **337** | **494** | **611** | **772** | **809** | **864** | **832** | **742** | **653** | **465** | **358** | (10) |
| (11) | | | CDD18.3 | **4222** | **60** | **109** | **237** | **361** | **514** | **559** | **606** | **573** | **492** | **394** | **215** | **102** | (11) |
| (12) | | | CDH23.3 | **65647** | **396** | **993** | **2797** | **5252** | **8644** | **9798** | **10803** | **9982** | **8268** | **5828** | **2164** | **722** | (12) |
| (13) | | | CDH26.7 | **44955** | **55** | **326** | **1377** | **3245** | **6214** | **7410** | **8324** | **7509** | **5886** | **3554** | **890** | **164** | (13) |
| (14) | **Wind** | | WSAvg | **3.6** | **3.4** | **4.0** | **3.9** | **4.1** | **4.0** | **4.1** | **3.8** | **3.4** | **3.2** | **2.9** | **2.8** | **3.3** | (14) |
| (15) | **Precipitation** | | PrecAvg | **122** | **16** | **19** | **25** | **14** | **5** | **3** | **7** | **6** | **4** | **4** | **6** | **12** | (15) |
| (16) | | | PrecMax | **285** | **44** | **78** | **156** | **75** | **15** | **15** | **48** | **19** | **18** | **22** | **32** | **58** | (16) |
| (17) | | | PrecMin | **48** | **1** | **1** | **0** | **0** | **0** | **0** | **1** | **1** | **0** | **0** | **0** | **1** | (17) |
| (18) | | | PrecStd | **47** | **11** | **16** | **31** | **18** | **4** | **4** | **9** | **4** | **3** | **5** | **8** | **15** | (18) |
| (19) | **Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures** | 0.4% | DB | **30.0** | **33.8** | **37.4** | **41.0** | **46.1** | **47.1** | **47.1** | **46.1** | **44.4** | **40.7** | **34.9** | **32.5** | | (19) |
| (20) | | | MCWB | **15.2** | **16.7** | **16.7** | **18.9** | **19.6** | **20.5** | **21.3** | **21.5** | **20.7** | **19.0** | **17.8** | **16.8** | | (20) |
| (21) | | 2% | DB | **28.1** | **32.0** | **35.7** | **39.7** | **43.7** | **46.0** | **46.0** | **44.9** | **43.0** | **39.4** | **33.4** | **29.7** | | (21) |
| (22) | | | MCWB | **15.3** | **16.5** | **16.5** | **18.5** | **19.7** | **20.3** | **21.4** | **21.8** | **20.7** | **18.9** | **17.9** | **15.8** | | (22) |
| (23) | | 5% | DB | **26.8** | **30.4** | **34.3** | **38.5** | **42.4** | **44.9** | **45.0** | **43.9** | **41.9** | **38.2** | **32.4** | **28.4** | | (23) |
| (24) | | | MCWB | **15.2** | **16.0** | **16.6** | **18.1** | **19.6** | **20.2** | **21.5** | **21.8** | **20.5** | **18.5** | **17.6** | **16.0** | | (24) |
| (25) | | 10% | DB | **25.6** | **28.7** | **32.9** | **37.3** | **41.2** | **43.7** | **44.0** | **42.9** | **40.8** | **36.9** | **31.2** | **27.1** | | (25) |
| (26) | | | MCWB | **14.8** | **15.7** | **16.4** | **17.7** | **19.4** | **20.3** | **21.5** | **21.8** | **20.5** | **18.3** | **17.3** | **15.7** | | (26) |
| (27) | **Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures** | 0.4% | WB | **18.2** | **20.3** | **21.4** | **22.7** | **24.5** | **26.0** | **26.9** | **26.3** | **24.9** | **23.5** | **21.2** | **19.7** | | (27) |
| (28) | | | MCDB | **24.8** | **27.0** | **27.9** | **29.3** | **33.7** | **33.6** | **36.3** | **36.9** | **35.1** | **31.1** | **28.1** | **25.1** | | (28) |
| (29) | | 2% | WB | **17.4** | **18.9** | **20.2** | **21.6** | **23.6** | **25.1** | **25.7** | **25.1** | **24.1** | **22.6** | **20.2** | **18.7** | | (29) |
| (30) | | | MCDB | **24.4** | **26.8** | **27.6** | **29.0** | **33.8** | **35.2** | **36.6** | **37.4** | **34.7** | **30.3** | **28.2** | **24.9** | | (30) |
| (31) | | 5% | WB | **16.5** | **18.0** | **19.3** | **20.7** | **22.6** | **24.4** | **25.0** | **24.6** | **23.6** | **21.7** | **19.5** | **17.9** | | (31) |
| (32) | | | MCDB | **24.0** | **26.3** | **28.3** | **30.6** | **34.2** | **35.4** | **36.8** | **37.0** | **34.5** | **31.0** | **27.9** | **24.3** | | (32) |
| (33) | | 10% | WB | **15.7** | **17.2** | **18.4** | **19.7** | **21.6** | **23.6** | **24.5** | **24.1** | **23.0** | **20.7** | **18.9** | **17.1** | | (33) |
| (34) | | | MCDB | **23.5** | **26.1** | **28.2** | **32.6** | **35.4** | **35.8** | **36.7** | **36.6** | **34.2** | **31.8** | **27.4** | **24.5** | | (34) |
| (35) | **Mean Daily Temperature Range** | | MDBR | **11.6** | **11.9** | **12.7** | **12.1** | **12.4** | **13.2** | **12.1** | **12.1** | **12.3** | **11.6** | **10.7** | **11.2** | | (35) |
| (36) | | 5% DB | MCDBR | **12.4** | **12.2** | **13.4** | **12.4** | **13.4** | **14.0** | **13.0** | **12.4** | **12.7** | **12.4** | **11.3** | **12.7** | | (36) |
| (37) | | | MCWBR | **4.9** | **4.7** | **4.8** | **3.9** | **4.2** | **4.3** | **3.8** | **3.4** | **3.7** | **4.2** | **4.2** | **4.6** | | (37) |
| (38) | | 5% WB | MCDBR | **10.8** | **10.1** | **10.3** | **10.3** | **11.5** | **13.1** | **11.9** | **12.0** | **12.4** | **11.1** | **10.5** | **10.2** | | (38) |
| (39) | | | MCWBR | **4.9** | **4.6** | **4.6** | **4.1** | **4.7** | **4.5** | **3.4** | **3.2** | **3.3** | **3.9** | **4.4** | **4.3** | | (39) |
| (40) | **Clear-Sky Solar Irradiance** | | taub | **0.397** | **0.445** | **0.506** | **0.562** | **0.584** | **0.661** | **0.748** | **0.708** | **0.594** | **0.470** | **0.415** | **0.392** | | (40) |
| (41) | | | taud | **2.179** | **2.015** | **1.847** | **1.752** | **1.708** | **1.587** | **1.487** | **1.546** | **1.721** | **1.977** | **2.144** | **2.190** | | (41) |
| (42) | | | Ebn at Noon | **855** | **835** | **802** | **765** | **744** | **683** | **626** | **651** | **719** | **800** | **828** | **843** | | (42) |
| (43) | | | Edh at Noon | **133** | **167** | **207** | **232** | **242** | **271** | **299** | **282** | **232** | **172** | **137** | **127** | | (43) |
| (44) | **All-Sky Solar Radiation** | | RadAvg | **4.70** | **5.48** | **6.28** | **6.92** | **7.59** | **7.53** | **6.93** | **6.84** | **6.48** | **5.90** | **4.95** | **4.46** | | (44) |
| (45) | | | RadStd | **0.21** | **0.24** | **0.34** | **0.41** | **0.19** | **0.25** | **0.26** | **0.17** | **0.13** | **0.15** | **0.10** | **0.18** | | (45) |

Note: in rows (19)–(45) the values occupy the Jan–Dec columns (e)–(p); the Annual column (d) is empty for these rows.

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (11 neighbors) | **+0.30** | **+0.52** | **N/A** | **N/A** | **+0.26** | **+0.29** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

[Nomenclature: See separate page]